



Application Note

Probe Temperature Sensing

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Summary

The Vantage system provides HW circuitry and SW utilities to allow monitoring of one or two transducer temperature sense thermistors connected through dedicated pins at the probe connector, for probes that provide this feature. The system can be programmed to automatically monitor the temperature sensors while a script is running, and exit the script with an error condition if a predetermined temperature limit is exceeded. The temperatures can also be read from the hardware system whenever desired, for display or other purposes. At the hardware level, the system simply measures the resistance of the device connected to the thermistor signals at the probe connector. If that device is a thermistor you can translate the measured resistance into a temperature, but you can also use this feature to measure the resistance of some other type of sensor or control within the probe. This document describes the functionality of the hardware feature, how to program it for monitoring and for enabling the automatic shutdown threshold, and how to convert the monitored value into a temperature.

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Document Change History

Date	Section(s)	Description	Author
April 13, 2017	All	Rev A: Initial document	KL

July 23, 2021		Updates for 4.5 software release	KL
Feb. 15, 2022	4	Add UTA-408 temperature monitoring support	KL

Description of the Hardware-Level Monitoring Function

The hardware system is designed to allow monitoring of thermistors (or other devices) through three dedicated signal pins at the probe connector interface: one for each of two sensors within the probe, and a return signal shared by both of them. Note that not all probe connectors supported by the Vantage system provide these signals, and within probe families that do provide the capability some probes may not use it. Refer to the UTA module documentation for a specific connector and probe type to determine whether the thermistor signals are provided and if so, which connector pins are used for the three signals.

This feature was originally designed to support the thermistors used in some ATL-Philips “HDI format” commercial transducers (using the “UTA 260” connector modules), but it can also be used to monitor the resistance of any other device within the transducer (such as a position sense potentiometer) as long as the resistance to be measured is in the range of approximately 100 to 10,000 Ohms. (Values well outside of this range may be difficult to measure with any precision.)

At the transducer connector, each thermistor or other resistance to be measured should be connected to one of the two sense signals, with the other end of both devices connected to the common return signal. The return signal is connected to ground within the system, but within the probe it should not connect to ground and all three of these sense signals should be electrically isolated from all other circuitry. This is to prevent any current flow through the thermistor signals from other devices within the probe, since any such current would corrupt the thermistor resistance measurement.

Within the Vantage hardware system, the two thermistor sense signals are each driven through a 2490 Ohm 0.1% resistor from a 2.5 Volt precision voltage reference source, and are monitored by a 12 bit A/D converter using the same 2.5 Volt reference. Thus when a transducer thermistor is connected between one of the sense signals and the common ground return, it will form a voltage divider with the 2490-Ohm resistor from the 2.5 Volt source, and the A/D will be monitoring the resulting voltage at the center of the divider. If the voltage being read by the A/D is V_{in} and the unknown thermistor resistance is R_x , we will have the following voltage divider relationship:

$$V_{in} = 2.5 \text{ V.} * R_x / (R_x + 2490)$$

The system software provides a function you can call from within the Matlab workspace that will return the 12 bit A/D value representing V_{in} , as an integer in the range 0 : 4095. If AD_v represents this value from the A/D, and noting that a 2.5 V. full-scale reading from the A/D would be 4095, we can solve for R_x :

$$R_x = AD_v * 2490 / (4095 - AD_v)$$

The accuracy of this measurement will be set by the 0.1% tolerance of the 2490 Ohm resistor, plus the linearity of the A/D. Note that since the A/D and the circuit being monitored share the 2.5 Volt reference source, any error in the reference voltage will not affect the result. The other potential source of inaccuracy you should be aware of and correct for is the resistance of the wiring between the thermistor (in the probe) and the A/D (inside the system). The resistance measurement as described above will include the wiring resistance that is in series with the thermistor; in a typical ultrasound probe assembly with very fine-gauge wires in the cable, this wiring resistance could be several Ohms or possibly even 10 to 20 Ohms. To get an accurate temperature reading, you should measure or estimate the wiring resistance, subtract that from the resistance derived from the A/D reading and then use the result as the input to the resistance vs temperature characteristic of the thermistor being used.

The hardware system obtains a new A/D sample value for each probe thermistor at intervals of approximately 2 msec when no active transmit-receive events are underway. If the time to start a new thermistor A/D conversion occurs while an acquisition event is in process, that A/D conversion will be delayed until the acquisition event is complete. This will reduce the net sample rate, but will insure that the thermistor reading is not corrupted by coupling from the very high transmit currents that may be present during acquisition. The software function used to query the HW system for the thermistor A/D values is “vsv.hal.shi.thermo.readProbeThermistorValues”; when a thermistor reading is requested through this function, the returned value will simply be the most recently acquired reading from the A/D.

UTA Modules / Connector Types Supporting Temperature Monitoring

Listed below are the UTA modules available as of the writing of this document that support the probe thermistor monitoring function, and for each of those modules the probe connector pins that are used.

- **UTA 260-S, UTA 260-D, UTA 260-MUX:** All three of these adapter modules use the Cannon DL-260 connector with a pinout compatible with the HDI-series probes from ATL/Philips. Temperature sensing is provided through connector pins cc4 and cc5 for sensors 1 and 2 respectively, and pin cc8 as the common sensor return. Pre-UTA Vantage systems with the built-in connector

are functionally equivalent to the UTA-260 adapter modules, including probe thermistor support.

- **UTA 408:** Cannon DLP408 connector with a pinout defined by Verasonics (does not support any commercial probes). Temperature sensing is provided through connector pins D1 and F1 for sensors 1 and 2 respectively, and pins C1 and E1 as the common sensor return.
- **UTA 408-GE:** Cannon DLP408 connector supporting the GE “D-series” probe family. This adapter provides thermistor temperature sensing for sensor 1 only, at connector pin L32 for sensor 1 and pin L33 for the sensor return.

Using the Over-temperature Automatic Shutdown Feature

From within a user’s “SetUp...” script you can enable the probe temperature monitoring through the structure `Resource.Parameters.ProbeThermistor`. This is a 1 X 2 Matlab Structure array, with a separate structure for each of the two possible thermistors in the probe. Each structure contains three fields, with the following meanings:

- ***enable***: a scalar Matlab double, used as a logical to enable the automatic error threshold monitoring feature. Set to 1 to enable monitoring for this thermistor, or set to zero to disable monitoring. Note that if error threshold monitoring is disabled, the A/D readings of the thermistor resistance are still being taken and you can still use the ***vsv.hal.shi.thermo.readProbeThermistorValues*** function (see section 3) to read the current value.
- ***threshold***: a scalar Matlab double, set to an integer value in the range from 0 to 4095 representing the threshold value from the 12-bit thermistor monitor A/D at which the over-temperature error condition will be triggered.
- ***reportOverThreshold***: a scalar Matlab double, used as a logical to specify which direction from the ***threshold*** value represents an error. If ***reportOverThreshold*** is set to 1, any A/D reading greater than ***threshold*** will trigger the error condition and stop the system; if set to 0 any reading less than ***threshold*** will trigger the

error. This allows the error-triggering feature to be used with both positive and negative temperature coefficient thermistors.

If you do not specify `Resource.Parameters.ProbeThermistor` in your script, VSX will create it and set all values to zero, disabling the error threshold monitoring. If you do specify it, make sure to specify all three values in both structures since VSX will not provide default (zero) values for any missing or empty fields.

Note that the threshold value is in units of the “raw” 12-bit A/D reading, not a temperature value and not an equivalent resistance value. To determine the threshold value for a desired temperature you must first convert that temperature to a resistance value using the characteristics of the specific thermistor being used, and then use the equations in section 1 to convert the resistance into the expected A/D reading for that resistance. Alternatively, you could empirically determine the value by putting the probe into a water bath held at the desired threshold temperature and use the A/D reading taken from the system as the threshold value to use.

Reading the Thermistor Monitor Value

If you want to read the current values from the probe thermistor monitor (for example, to display the current temperature while the script is running or as input to your own mechanism for issuing over-temperature warnings or fault conditions), you can use the ***readProbeThermistorValues*** function described previously. The function call syntax is:

thermistorValues = vsv.hal.shi.thermo.readProbeThermistorValues(conn, numTH, VDAS)

Input argument definitions:

- ***conn*** - Matlab double. Selects which transducer connector is to be monitored. Set to desired connector index on a multi-connector UTA; always set to 1 for any system/UTA configuration with only one connector.
- ***numTH*** - Matlab double. Specifies the number of thermistors being used (1 or 2; the number of elements in `Resource.Parameters.ProbeThermistor`)

- **VDAS** - Matlab double set to 1 if using HW system, or 0 for simulation with no hardware.

Output argument definition:

- ***thermistorValues*** – Matlab doubles indicating the A/D reading from each thermistor being queried (integer value in range 0:4095).

The example script “SetUpL11_5vThermistorDemo” accompanying this application note provides a simplistic example of the use of the thermistor monitoring features. This script reads the thermistor monitor values from both sensors in an external processing function called from within the event sequence. This function creates a Matlab figure window and then uses it to display the measured A/D sample values from both thermistors as well as the equivalent resistance in Ohms that is calculated from the A/D sample value (as explained in section 1 of this document).

As shown in the example script, you can call the function ***convertThermistorValues*** to convert the thermistorValues as read from the A/D into actual resistance values as explained at the beginning of this document:

resistances = vsv.hal.shi.thermo.convertThermistorValues(ThermistorValues)

If you run the script with an actual L11-5v probe, an open circuit condition will be reported for both temperature sensors because the L11-5v does not have thermistors. You can run the script with no probe connected, and then experiment with connecting a resistor (or an actual thermistor) to the thermistor pins at the connector and see the actual measured value on the display.

You can also enable monitoring by setting non-zero values in the structure Resource.Parameters.ProbeThermistor near the beginning of the script, to demonstrate the fault condition and error message when the monitor threshold is exceeded.

Usage Guidelines, for User-defined Custom Probes

The bias current used to measure the thermistor resistance is intentionally kept very low (less than one milliamp), to minimize self-heating of the thermistor due to the bias current that would corrupt the temperature measurement. As a result, care should be taken to avoid any ground loops or other coupling paths that would couple noise into the thermistor wiring carrying this very low-level signal.

Note that the monitoring circuitry is measuring the resistance as seen within the Vantage system, at the connections to the bias resistor and the A/D converter. There may be several Ohms, or

even tens of Ohms, of resistance in series with the thermistor due to the fine-gauge wire used in typical ultrasound probe cable assemblies as well as PCB traces and the various connectors the signals must traverse. To get the most accurate temperature measurement possible, this additive resistance should be estimated or measured so it can be subtracted from the measured total resistance before applying the thermistor's resistance vs temperature characteristic. Another concept for even greater accuracy would be to calibrate each individual probe assembly by measuring the actual resistance while the probe is immersed in a water bath at a precisely regulated temperature. This measured calibration resistance at a known temperature could then be stored in the ID eeprom of the transducer, so the system SW could use it as reference point when estimating temperatures from that specific probe.

Another potentially significant source of error is the various heat flow paths that will exist within the physical probe assembly, between the active elements, thermistor, transducer face, and heat flow to the subject and the operator from the transducer housing. To get a meaningful estimate of the difference between the temperature at the thermistor and at the point of interest, these heat flow paths must be well characterized including potential variations from unit to unit. Response times due to the thermal mass of the various components of the probe assembly must also be considered, especially for short transmit burst durations at low repetition rates. Low pass filtering of the thermistor resistance measurements may also be needed, to reduce the effects of noise or other signals coupling into the thermistor wiring. It is critical that there be no DC current paths from the thermistor wiring within the probe assembly to anything else, including any other ground connections.

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